Jacobian Analysis of a Recurrent Transformer Block

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Abstract

We study the Jacobian of a recurrent Transformer update containing multi-head self-attention, a feed-forward network, residual connections, and RMSNorm. Applying the chain rule gives a bound that depends on the component Jacobians, the three normalization scales, and the smallest normalization denominator across tokens. For a fixed state, context, and set of weights, the one-step Jacobian remains bounded as the step coefficient tends to infinity, provided that every token's update is nonzero. We state the result for the standard RMSNorm convention with a numerical stabilizer and compare it with numerical Jacobians. Separate trajectory experiments on Gaussian inputs and projected CIFAR-10 images measure successive-state differences. Boundedness with respect to step size does not establish contraction, convergence, or bounded gradients through arbitrarily many recurrent steps.

Keywords: recurrent Transformers, Jacobian bounds, RMSNorm, recurrent dynamics.

1 Introduction

Self-attention is the central operation in Transformer blocks [31]. Its dynamics have been studied through rank collapse [24], approximation properties [35, 15, 23], and energy functions [28, 33, 14]. Reusing a block across iterations makes the distinction between a single update and a long trajectory particularly relevant.

Tomihari and Karakida [30] bound the Jacobian of recurrent self-attention with normalization. Their analysis also gives a bounded large-step limit under a non-degeneracy assumption. We apply the same argument to a block containing both attention and a feed-forward network, with two inner normalizations and an outer normalization. The resulting estimate makes the contribution of each component explicit. Adding an FFN and its residual path introduces another derivative factor and another normalization denominator.

The scope of the result is the norm of a single-step derivative. A bound of order $O(1)$ as the step coefficient grows can still be greater than one. It therefore does not, by itself, imply stability of a fixed point or convergence of a recurrent trajectory. We keep these questions separate in the analysis and experiments.

2 Related Work

Energy-based analysis. Mean-field and particle descriptions of self-attention explain clustering and metastability under specified architectural and dynamical assumptions [10, 11, 9, 4, 16]. Other work relates attention and Transformer updates to energy minimization [28, 33, 14]. The Jacobian approach used here does not require an energy function.

Jacobian-based analysis. Jacobians are used to study neural dynamics and signal propagation [3, 27, 25, 13, 6] and to motivate regularization [34, 21]. For attention, Noci et al. [24] analyze signal propagation, and Castin et al. [5] derive Lipschitz bounds. The closest result is the normalization bound of Tomihari and Karakida [30]. Our additional step is its composition through the two residual sub-blocks and their separate normalizations.

Loss-landscape curvature. Petrov et al. [26] analyze the Hessian of full Transformer blocks, including LayerNorm, feed-forward networks, and residual connections. Kiselev and Grabovoy [17] study changes in the loss landscape as the sample size grows. Their later work [18] introduces curvature-aligned probes of local landscape stabilization under sample growth. These studies concern second-order optimization geometry; the Jacobian considered here differentiates a recurrent state update while holding the model parameters fixed.

Looped architectures. Universal Transformers [7] reuse parameters across computational steps. Further work studies algorithmic learning, reasoning, and test-time computation in looped models [32, 12, 29, 8, 2]. Related constructions include ALBERT [20], deep equilibrium models [1], and oscillatory neurons [22]. We analyze one particular post-normalization update; the results do not cover all of these architectures.

3 Methodology

3.1 Notation and preliminaries

The state is $Y \in \mathbb{R}^{S\times D}$, where S is sequence length and D is hidden dimension. We write $y_i = Y_{[i,:]}^\top$ when a token is treated as a column vector. A full-state Jacobian acts on the row-wise vectorization of Y and has size $SD \times SD$. Its spectral norm is the operator norm induced by the Frobenius norm on states. Vector norms are Euclidean unless specified otherwise. The context C is held fixed when differentiating with respect to Y.

3.2 Basic building blocks

Multi-head self-attention. For H heads and $d_h = D/H$, let

$$Q = YW_Q + \mathbf{1}b_Q^\top, \quad K = YW_K + \mathbf{1}b_K^\top, \quad V = YW_V + \mathbf{1}b_V^\top. \tag{3.1}$$

Splitting the last dimension into H contiguous blocks gives Q_h, K_h, V_h. The attention operation is

$$A_h = \mathrm{softmax}\Big(Q_h K_h^\top/\sqrt{d_h}\Big), \qquad \mathrm{MSA}(Y) = [A_1V_1\|\cdots\|A_HV_H]W_O + \mathbf{1}b_O^\top, \tag{3.2}$$

where softmax acts row-wise. The implementation uses unmasked attention without dropout or positional encodings. Biases are included to match the implemented affine projections; setting them to zero gives the bias-free formulation.

Feed-forward network. The position-wise network is

$$\mathrm{FFN}(Y) = \sigma(YW_1 + \mathbf{1}b_1^\top)W_2 + \mathbf{1}b_2^\top, \tag{3.3}$$

where $W_1 \in \mathbb{R}^{D\times D_{\mathrm{ff}}}$, $W_2 \in \mathbb{R}^{D_{\mathrm{ff}}\times D}$, and σ is differentiable; the experiments use GELU. The bound below uses its local Jacobian norm, not an assumed Lipschitz constant of one.

RMSNorm. For each of the three normalization layers $\ell \in \{1, 2, U\}$, define

$$\rho_\ell(y) = \sqrt{\|y\|_2^2/D + \epsilon_\ell}, \qquad R_\ell(y) = \mathrm{diag}(\gamma_\ell)\frac{y}{\rho_\ell(y)}, \qquad g_\ell = \|\gamma_\ell\|_\infty, \tag{3.4}$$

with $\epsilon_\ell > 0$. We use R_ℓ row-wise on matrices. This is the usual RMS scaling [36], including a numerical stabilizer. When $\epsilon_\ell = 0$, it equals $\sqrt{D}\ \mathrm{diag}(\gamma_\ell)y/\|y\|_2$. Thus a unit-$L^2$ normalization uses a different scale convention; the two conventions must not share numerical scale parameters without conversion.

3.3 Iterative Transformer block

Define the internal states and the block output by

$$Z_1 = Y + \mathrm{MSA}(Y), \qquad Y' = R_1(Z_1), \tag{3.5}$$

$$Z_2 = Y' + \mathrm{FFN}(Y'), \qquad \mathrm{TB}(Y) = R_2(Z_2). \tag{3.6}$$

For a step coefficient $\nu > 0$, the recurrent update is

$$\Delta(Y) = C + \mathrm{TB}(Y), \qquad U_\nu = Y + \nu\Delta(Y), \qquad Y^{(t+1)} = F_\nu(Y^{(t)}) = R_U(U_\nu). \tag{3.7}$$

Weights and context are reused over the trajectory. This is an explicit update: no equation for $Y^{(t+1)}$ needs to be solved implicitly.

3.4 Analytic Jacobian bounds

3.4.1 Composed bound

At a given state, set

$$r_1 = \min_i \rho_1((Z_1)_{[i,:]}^\top), \qquad r_2 = \min_i \rho_2((Z_2)_{[i,:]}^\top), \qquad r_U = \min_i \rho_U((U_\nu)_{[i,:]}^\top), \tag{3.8}$$

$$M = \|J_{\mathrm{MSA}}(Y)\|_2, \qquad L = \left\|J_{\mathrm{FFN}}(Y')\right\|_2. \tag{3.9}$$

These are the smallest denominators across tokens, not their means.

Theorem 3.1 (One-step bound)**.** *For the map in Eq.* (3.7)*, with finite parameters and the positive stabilizers in Eq.* (3.4)*,*

$$\|J_{F_\nu}(Y)\|_2 \le \frac{g_U}{r_U}\left[1 + \nu\frac{g_2 g_1}{r_2 r_1}(1+L)(1+M)\right]. \tag{3.10}$$

The same expression holds if each r_ℓ is replaced by a positive lower bound for the corresponding minimum. With positive ϵ_ℓ, the exact minima are always positive; small minima can nevertheless make the estimate large. No claim that all normalization factors are less than one is needed. The proof is in Appendix A.2.

3.4.2 Large-step limit

Theorem 3.2 (Boundedness with respect to step size)**.** *Fix Y, C, all weights, and the normalization parameters independently of ν. Suppose*

$$\delta = \min_i \left\|\Delta(Y)_{[i,:]}\right\|_2 > 0. \tag{3.11}$$

Let $B = \max_i \left\|Y_{[i,:]}\right\|_2$ and $K = g_2 g_1 (1+L)(1+M)/(r_2 r_1)$. For $\nu \geq 2B/\delta$ and $\nu > 0$,

$$\|J_{F_\nu}(Y)\|_2 \leq \frac{2\sqrt{D}\, g_U}{\delta}\left(K + \frac{1}{\nu}\right). \tag{3.12}$$

Consequently $\|J_{F_\nu}(Y)\|_2 = O(1)$ as $\nu \to \infty$.

The constants may depend on the fixed state and parameters. A statement uniform over states or trajectories would additionally require uniform bounds on B, K, and a uniform positive lower bound on δ. The proof in Appendix A.3 adapts the large-step argument of [30].

3.5 What the bounds do not imply

For T recurrent steps, the state derivative is

$$\frac{\partial Y^{(T)}}{\partial Y^{(0)}} = J_{F_\nu}(Y^{(T-1)}) \cdots J_{F_\nu}(Y^{(0)}), \tag{3.13}$$

when C is independent of the differentiated initial state. Bounding every factor by a constant $k > 1$ only gives the bound k^T for the product. If C is itself a function of the initial state, additional context derivative terms appear. Theorem 3.2 controls neither case uniformly in T.

At a differentiable fixed point, spectral radius less than one is a sufficient condition for local asymptotic stability. A spectral norm below one is a stronger sufficient condition. A uniform derivative norm bound below one on an appropriate invariant convex region yields a contraction argument. None of these conditions follows from $O(1)$ in ν. For example, the smooth map $x \mapsto -x$ on a sphere has bounded Jacobian norm and period-two trajectories. This elementary example illustrates the logical distinction; it is not an experimental result about trained Transformers.

4 Experimental Results

The experiments use randomly initialized, untrained blocks, with $\gamma_1 = \gamma_2 = \gamma_U = \mathbf{1}$ and $\epsilon_1 = \epsilon_2 = \epsilon_U = 10^{-8}$. They test numerical agreement with the bound and illustrate selected trajectories. They do not evaluate task accuracy or establish behavior after training. Full-state and component Jacobian norms in the first two experiments are obtained from dense automatic differentiation and singular value decomposition (SVD), using float64.

4.1 Validation of the one-step bound

We use $S = 16$, $D = 32$, $H = 4$, and $D_{\text{ff}} = 64$, with ten equally spaced values of ν from 0.1 to 2.0. The model, Gaussian state $Y_{ij} \sim \mathcal{N}(0, 0.5^2)$, and independent context $C_{ij} \sim \mathcal{N}(0, 0.1^2)$ are sampled once with seed 42 and held fixed throughout the sweep. We compute M, L, and the three minima in Eqs. (3.8)–(3.9), and compare the resulting bound with $\|J_{F_\nu}(Y)\|_2$. The measured component norms are $M = 0.9607$ and $L = 0.5486$. The inequality holds at all ten sampled step sizes. The one-step norm increases from 2.6713 at $\nu = 0.1$ to 3.6547 at $\nu = 2.0$; the corresponding bounds are 3.5924 and 7.0763. These values illustrate a conservative bound and do not certify contraction, since the measured norms exceed one.

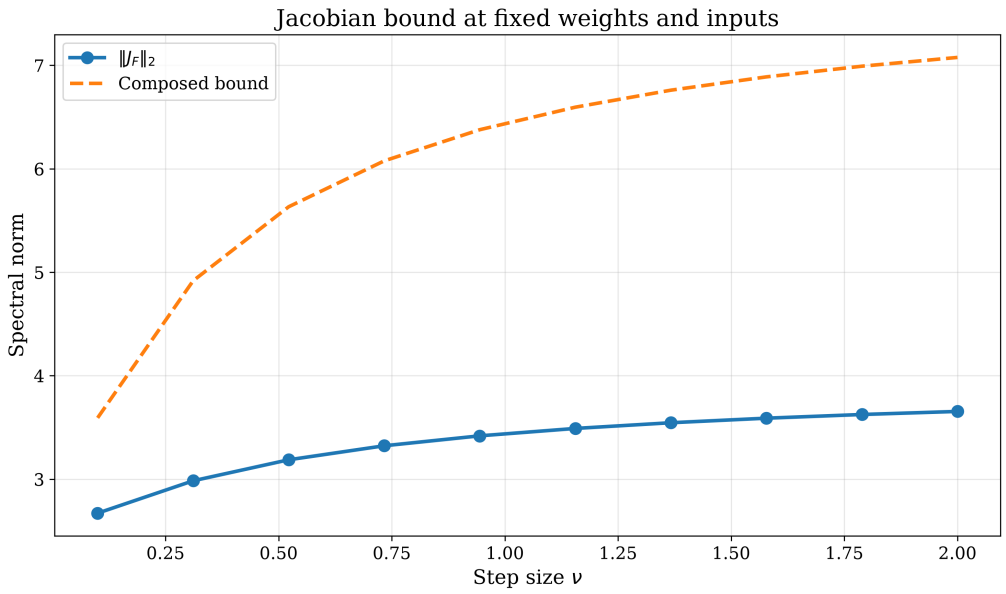


Figure 1: One-step Jacobian norm and the composed bound for a fixed model, state, and context. Component norms are computed at the same states used by the recurrent update.

4.2 Large-step behavior

We use the same dimensions and Gaussian distributions, with seeds 0 through 9. Each seed specifies a model, state, and context that remain fixed across twenty logarithmically spaced values of ν from 10^{-1} to 10^{5}. Means and standard deviations are taken across these ten independent repetitions, rather than across inputs to a single model newly sampled at each step size. This protocol follows the fixed-parameter scope of Theorem 3.2. We also compute the Jacobian of the limiting map

$$(G_\infty(Y))_i = \sqrt{D}\,\operatorname{diag}(\gamma_U)\frac{\Delta(Y)_i}{\|\Delta(Y)_i\|_2}. \tag{4.1}$$

Here tokens are written as column vectors. The positive lower bound on the update norms makes this map differentiable in a neighborhood of the sampled state. The outer stabilizer vanishes in the limit because it is divided by ν^2; both inner stabilizers remain part of TB. The reference line in Figure 2 is the mean of $\|D_Y G_\infty(Y)\|_2$ across seeds, computed separately from the finite-step curves. It is not the norm of a mean Jacobian or the last sampled ordinate. The mean norm is 2.8079 at $\nu = 0.1$ and 3.720649 at $\nu = 10^5$, compared with a mean limiting norm of 3.720655. At the largest step, the standard deviation across seeds is 0.1538, and the largest per-seed absolute difference from the corresponding limiting norm is 7.94×10^{-6}. The smallest update-row norm across the ten runs is 5.4294, so all sampled states satisfy the non-degeneracy condition. The finite-step curves are consistent with the large-step limit; their plateau above one does not imply contraction.

4.3 Successive-state differences

For the trajectory experiment we use $S = 32$, $D = 64$, $H = 4$, and $D_{\text{ff}} = 128$. Starting from $Y^{(0)}$, we hold $C = 0.1Y^{(0)}$ fixed and record

$$d_t = \left\|Y^{(t+1)} - Y^{(t)}\right\|_F \tag{4.2}$$

over 1000 steps for $\nu \in \{0.1, 0.5, 1.0\}$. Inputs are Gaussian sequences or CIFAR-10 test images [19] mapped to sequences using a random linear projection. The projection and the recurrent block

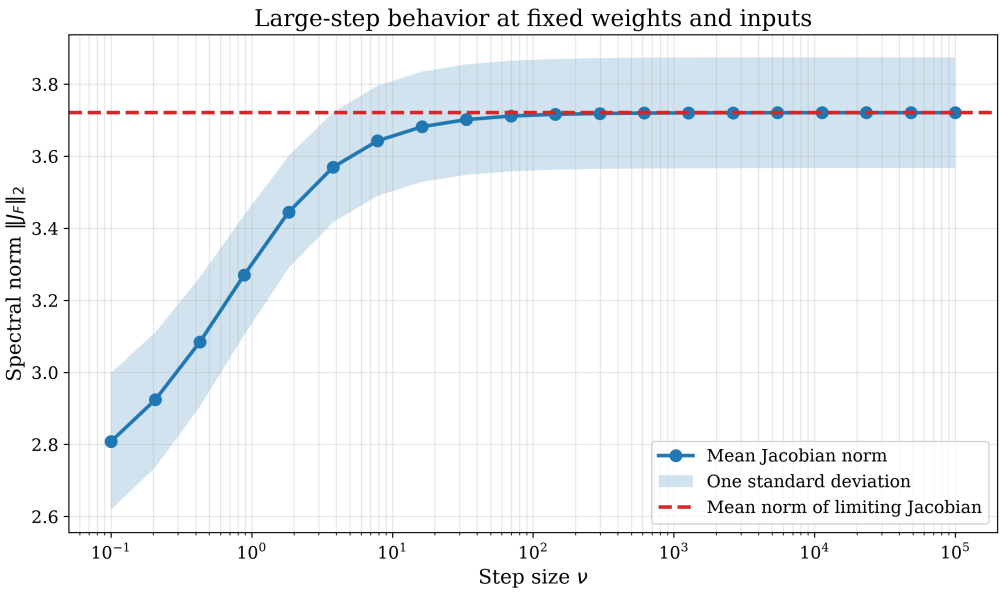


Figure 2: Dependence of the one-step Jacobian norm on step coefficient. The horizontal axis is logarithmic. The curve is the mean and the band is one population standard deviation across ten seeds; within each seed only ν changes. The dashed reference is the mean norm of the limiting Jacobian.

are untrained. We use seeds 0 through 4, with five Gaussian samples and the five fixed CIFAR-10 test images at indices 0 through 4 for each seed, giving 25 trajectories per dataset and step size. Each image is scaled to $[-1, 1]$, divided by its own sample standard deviation, reshaped into 32 rows of 96 values, and projected to dimension 64. For each seed, the same block weights are used for both datasets and all step sizes. Model weights, Gaussian inputs, and projection weights use separate random-number streams.

We report the median residual and the observed minimum-to-maximum range. The finite-run criterion is $d_t \leq 10^{-4}$ throughout the final 20 steps; passing it is not a claim of asymptotic convergence. The trajectories use float32, so late residuals also depend on numerical precision. Table 1 gives the number of trajectories meeting the residual criterion. Some sampled trajectories have not met the threshold after 1000 steps. This does not show that they fail to converge as $t \to \infty$. For this sample and iteration budget, larger ν increases the fraction meeting the criterion; this is an observation about the tested conditions, not a general monotonicity result.

Table 1: Trajectories with $d_t \leq 10^{-4}$ throughout the final 20 of 1000 steps. Each entry is a count out of 25 trajectories.

Input	$\nu = 0.1$	$\nu = 0.5$	$\nu = 1.0$
Gaussian	6/25	23/25	25/25
CIFAR-10	6/25	20/25	22/25

Zero residuals are displayed at 10^{-12} on the logarithmic plot; the saved arrays retain their original values. A decrease in d_t over a finite run is evidence about those trajectories only. Even the asymptotic condition $d_t \to 0$ alone is not a general proof that a sequence converges. The experiment therefore supplies no global convergence guarantee.

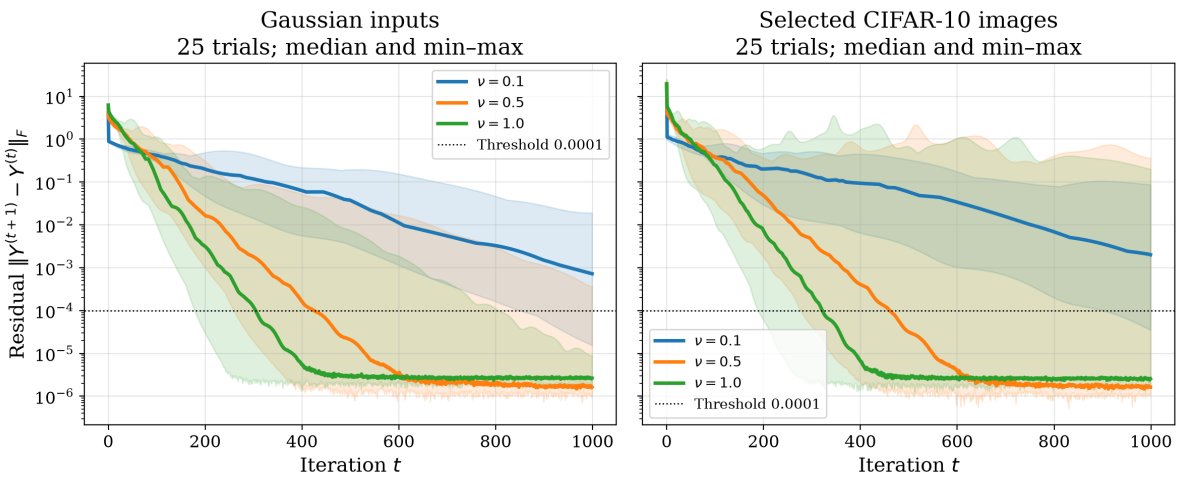


Figure 3: Successive-state differences on Gaussian inputs and projected CIFAR-10 images. Lines show medians across 25 trajectories; shaded bands span the observed minimum and maximum, not confidence intervals. The dotted line marks the 10^{-4} residual threshold.

5 Conclusion

The chain rule extends the recurrent self-attention Jacobian bound to a block with a feed-forward network and two inner RMSNorm layers. The estimate uses separate normalization scales and the smallest RMS denominator at each layer. For fixed inputs and parameters, a nonzero per-token update makes the outer denominator grow with ν, compensating for the explicit step multiplier. This establishes boundedness of the one-step Jacobian in the large-step limit. Contraction, stability over many steps, and behavior of trained networks require additional conditions or measurements.

A Proofs

A.1 Derivative of RMSNorm

Lemma A.1. *For* $R(y) = \mathrm{diag}(\gamma)y/\rho(y)$ *with* $\rho(y) = \sqrt{\|y\|_2^2/D + \epsilon}$ *and* $\epsilon > 0$,

$$J_R(y) = \frac{\mathrm{diag}(\gamma)}{\rho(y)}\left(I - \frac{yy^\top}{D\rho(y)^2}\right), \qquad \|J_R(y)\|_2 \le \frac{\|\gamma\|_\infty}{\rho(y)}. \tag{A.1}$$

Proof. Differentiation gives $\nabla\rho(y) = y/(D\rho(y))$, and the product rule gives Eq. (A.1). The matrix in parentheses has eigenvalue one in directions orthogonal to y, and eigenvalue $\epsilon/\rho(y)^2$ in the direction of y. Both lie in $[0, 1]$. At $y = 0$ the matrix is the identity. Submultiplicativity yields the stated inequality. Row-wise normalization has a block-diagonal Jacobian, so its norm is bounded by the maximum token bound. □

For $\epsilon = 0$ and nonzero input, the matrix in parentheses is the tangent projector. This recovers Lemma A.3 of [30], after accounting for the $\sqrt{D}$ scale convention. At the outer layer the derivative is evaluated at U_ν, not at the incoming state Y; with positive ϵ the matrix is no longer an exact projector.

A.2 Proof of Theorem 3.1

The chain rule gives

$$J_{\mathrm{TB}}(Y) = J_{R_2}(Z_2)(I + J_{\mathrm{FFN}}(Y'))J_{R_1}(Z_1)(I + J_{\mathrm{MSA}}(Y)), \tag{A.2}$$

$$J_{F_\nu}(Y) = J_{R_U}(U_\nu)(I + \nu J_{\mathrm{TB}}(Y)). \tag{A.3}$$

Lemma A.1 and the block-diagonal norm identity imply

$$\|J_{\mathrm{TB}}(Y)\|_2 \le \frac{g_2}{r_2}(1+L)\frac{g_1}{r_1}(1+M) = K. \tag{A.4}$$

Substitution into $\|J_{F_\nu}(Y)\|_2 \le (g_U/r_U)(1 + \nu\,\|J_{\mathrm{TB}}(Y)\|_2)$ proves Eq. (3.10). □

A.3 Proof of Theorem 3.2

For every token, the reverse triangle inequality gives

$$\left\|(U_\nu)_{[i,:]}\right\|_2 \ge \nu\left\|\Delta(Y)_{[i,:]}\right\|_2 - \left\|Y_{[i,:]}\right\|_2 \ge \nu\delta - B. \tag{A.5}$$

For $\nu \ge 2B/\delta$ this is at least $\nu\delta/2$. Consequently

$$r_U = \min_i \sqrt{\left\|(U_\nu)_{[i,:]}\right\|_2^2/D + \epsilon_U} \ge \frac{\nu\delta}{2\sqrt{D}}. \tag{A.6}$$

The inner states and K are independent of ν because Y, C, and all parameters are fixed. Theorem 3.1 now yields

$$\|J_{F_\nu}(Y)\|_2 \le \frac{2\sqrt{D}\,g_U}{\nu\delta}(1+\nu K), \tag{A.7}$$

which is Eq. (3.12). This argument does not assume that Y was produced by a preceding normalization. □

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